EduVision DV: Higher Education Performance Dashboard

Milestone 2 & 4: Dashboard Storyboard, Layout Blueprint & Technical Specifications

Executive Summary:
The EduVision DV analytics suite transforms complex global higher education metrics from QS World University Rankings and Times Higher Education (THE) into an executive-ready Power BI reporting platform. The suite features four interconnected, highly responsive analytical views designed for academic leadership, institutional researchers, policy analysts, and prospective students.

Dashboard View	Primary Target Audience	Core Analytical Objective	Key Visual Deliverables
1. University Overview	Provosts, Presidents, Global Strategy Teams	Executive benchmarking, institutional standing, macro regional distributions, and top-tier score evolutions.	6 KPI Cards, Global Rank Top 10, Score Trend (2020-2024), Regional Donut, Top 5 Pubs, Map, Faculty Ratio.
2. Research Analytics	VPs of Research, Grant Officers, Deans	Publication productivity, citation impact, researcher output elasticity, and discipline benchmarking.	Research Impact vs Volume Quadrants, Citations/Faculty Scatter, Subject Area Treemap, Top Research Hubs.
3. Student Analytics	Deans of Admissions, Student Affairs, HR	Internationalization percentages, faculty-to-student ratios, diversity distribution, enrollment capacity.	Global Student Heatmap, Faculty-to-Student Ratio Bar Chart, International Student %, Enrollment Trends.
4. Country Comparison	Government Ministries, Policy Analysts	Sovereign higher-ed capacity, regional excellence concentration, top-100 density, cross-border competitiveness.	Country Benchmark Matrix, Sovereign Score Radar, Regional Distribution Treemap, Multi-Country Ranking.

Design System & UI Color Tokens (High-Contrast Modern Dark Theme)

Canvas Canvas: #0B0E1B Deep Space Navy	**Surface Card:** #1B1F38 Elevation Fill	**Primary Accent:** #7C5CFC Electric Indigo / Violet	**Secondary Accent:** #3B82F6 Vibrant Azure Blue
Cyan Accent: #06B6D4 High-Impact Teal	**Positive KPI:** #10B981 Emerald Growth Indicator	**Warning/Highlight:** #F59E0B Amber Benchmark Marker	**Typography:** Segoe UI / Inter #FFFFFF & #94A3B8 Secondary

Page 1: University Overview Dashboard (Wireframe & Visual Architecture)

Exact match to design layout requirements: Dark Canvas (1920x1080 / 16:9), Header Slicers, Left Nav, 6 KPI Cards, and 6 Deep Visuals

#1 MIT Top Global Rank	1,503 Total Ranked Univs	72.6 ▲+2.4% Avg Overall Score	28.7% ▲+1.8% Intl Students %	1 : 17.3 ▼-0.6 Faculty Ratio	2.45M ▲+6.3% Total Publications

LEFT NAV & FILTERS • [Overview (Active)] • [Research Analytics] • [Student Analytics] • [Country Comparison] *Slicers:* Year (2024) Region (All) Country (All) Subject (All) **[Reset Filters]**	**VISUAL 1: Top 10 Universities by Global Ranking** Columns: Rank | University | Country | Score 1. MIT (USA) - 100.0 2. Cambridge (UK) - 98.7 3. Oxford (UK) - 97.9 4. Harvard (USA) - 97.1 5. Stanford (USA) - 96.8 6. Imperial (UK) - 95.4 7. ETH Zurich (CH) - 93.7 8. NUS (SG) - 92.9 9. UCL (UK) - 91.8 10. UC Berkeley (USA) - 90.8	**VISUAL 2: Top 10 Univs Score Trend (2020-2024)** Multi-Line Chart: • X-Axis: 2020, 2021, 2022, 2023, 2024 • Y-Axis: Overall Score (80 - 100) • Lines: MIT (#7C5CFC), Cambridge (#3B82F6), Oxford (#06B6D4), Harvard (#EC4899), Stanford (#F59E0B) • Cross-highlight on click	**VISUAL 3: Universities by Region** Donut Chart (Total: 1,503): • Europe: 37.6% (565) • North America: 28.3% (425) • Asia: 24.8% (373) • Oceania: 6.1% (92) • South America: 3.2% (48)
DATA FOOTER: Source: QS & THE 2024 Updated: Oct 2026	**VISUAL 4: Publications by Top 5 Universities** Horizontal Bar Chart: • Harvard: 208K • Stanford: 189K • MIT: 186K • Oxford: 163K • Cambridge: 150K	**VISUAL 5: International Students % by Region** Filled Global Map / Choropleth: Color gradient: 10% (Dark Blue) to 45% (Bright Orange) Oceania (39.0%), Europe (32.0%), N. America (21.5%), Asia (16.5%), S. America (6.5%)	**VISUAL 6: Faculty to Student Ratio by Region** Clustered Column Chart (Students per Faculty): • North America: 1:25.0 • Europe: 1:21.4 • Asia: 1:18.7 • Oceania: 1:16.2 • South America: 1:14.3

Page 2: Research Analytics Dashboard (Wireframe & Visual Architecture)

Focus: Publications output, Citation velocity, Research impact vs productivity quadrants, and Discipline breakdown

2.45M	12.8M	71.6	5.2	Harvard University	Caltech (100.0)
Total Indexed Publications	Cumulative Citations	Avg Research Impact Score	Avg Citations / Publication	Top Volume Leader (208K)	Top Citation Impact Leader

VISUAL 1: Research Impact vs Publications (Scatter Plot)
• X-Axis: Publications Volume (0 - 250K)
• Y-Axis: Research Impact Score (0 - 100)
• Bubble Size: Total Faculty Count
• Bubble Color: Continental Region
• Quadrant Lines: Median Publications & Impact
Highlights elite powerhouses in upper-right quadrant.

VISUAL 2: Research Output by Subject Area (Treemap)
• Discipline breakdown:
- Engineering & Technology (32%)
- Life Sciences & Medicine (28%)
- Natural Sciences (22%)
- Social Sciences & Management (12%)
- Arts & Humanities (6%)
• Tooltip reveals average citations per faculty.

VISUAL 3: Top 15 Research Universities (Horizontal Stacked Bar)
• Institutional ranking showing Publications volume breakdown by Discipline.
• Direct institutional filtering and cross-highlighting on click.
• Top 5: Harvard, Toronto, Stanford, MIT, Tsinghua.

VISUAL 4: Research Productivity Index Trend (2020-2024)
• Area / Line chart showing research productivity (publications per faculty) compound annual growth rate (CAGR).
• Demonstrates regional acceleration in East Asia and stable efficiency in Western Europe.

Page 3: Student Analytics Dashboard (Wireframe & Visual Architecture)

Focus: Student diversity, international student mobility, faculty-to-student ratios, and institutional capacity

32.8M	7.89M	28.7%	1 : 17.3	Imperial College (60.2%)	Caltech (1 : 1.9)
Total Student Population	Total International Students	Avg International Ratio	Global Avg Faculty Ratio	Highest International %	Best Faculty-Student Ratio

VISUAL 1: International Student Concentration Map
• Bubble Map showing destination intensity for international scholars.
• UK and Australian universities exhibit >40% average international cohort.
• Slicer allows filtering by study discipline.

VISUAL 2: Faculty-to-Student Ratio Distribution (Histogram / Bar)
• Distribution curve showing institutional staffing density.
• Bins: 1:1-10 (Elite research), 1:10-20 (Standard high-tier), 1:20-30 (Large state universities), >1:30 (Mass enrollment).

VISUAL 3: International Cohort vs Total Enrollment (Scatter)
• X-Axis: Total Student Headcount (Log scale: 1K - 100K)
• Y-Axis: International Students % (0% - 70%)
• Reveals institutional internationalization strategies across boutique vs mega-universities.

VISUAL 4: Student Body Growth & International % Trend (2020-2024)
• Dual-Axis Line & Clustered Column:
- Primary Axis: Total Headcount Growth (Columns)
- Secondary Axis: International Student % (Line with markers)
• Tracks post-pandemic international mobility recovery.

Page 4: Country Comparison Dashboard (Wireframe & Visual Architecture)

Focus: Cross-national performance benchmarking, top-100 density, regional excellence, and sovereign capacity

38	United States (28)	United Kingdom (17)	Switzerland (84.2)	Australia (41.8%)	Singapore (90.2)
Sovereign Nations Evaluated	Most Top 100 Univs	2nd Most Top 100 Univs	Highest National Avg Score	Highest Avg Intl Ratio	Highest Academic Rep Avg

VISUAL 1: Country Benchmark Matrix (Matrix / Heatmap Table)
• Rows: Sovereign Country (USA, UK, Australia, China, Germany, Switzerland...)
• Values: Total Universities, Top 100 Count, Avg Score, Avg Citations, Avg Intl %, Faculty Ratio.
• Conditional data bars and background color heatmaps.

VISUAL 2: Regional Performance Radar / Grouped Bar Comparison
• Multi-metric comparison across Europe, North America, Asia, Oceania, and South America.
• Compares Overall Score, Research Impact, Internationalization, and Faculty Density on a common 0-100 scale.

VISUAL 3: Top 100 Concentration by Country (Treemap / Bar)
• Visualizes geographic clustering of world-class universities.
• USA (28%), UK (17%), Australia (9%), China (7%), Germany (6%), Switzerland (4%), Canada (4%).
• Click-to-filter dynamically controls surrounding visuals.

VISUAL 4: Country-level Internationalization vs Research Impact
• Bubble chart plotting Sovereign Avg Intl Student % (X-axis) vs Sovereign Avg Research Impact Score (Y-axis).
• Identifies global models: Highly internationalized UK/Australia vs Volume-driven East Asian leaders.

Technical Specifications: Navigation, Slicers, Bookmarks & DAX Model

End-to-End Enterprise Power BI Implementation Blueprint

Feature Area	Power BI Implementation Mechanism	User Interaction & Behavior
Cross-Page Navigation	Left vertical navigation rail with 4 button objects configured with Action = Page Navigation.	Instant single-click switching between Overview, Research, Student, and Country views while preserving slicer state.
Slicer Synchronization	Sync Slicers pane active for Year, Region, Country, and Subject Area across all 4 pages.	Selecting 'Europe' on Overview persists across Research, Student, and Country views without manual reselection.
Reset Filters	Bookmark created with Default State (All filters cleared) tied to a 'Reset Filters' icon button.	Restores all 4 global slicers to default aggregated view across all visual cards.
Drill-Through Actions	Drill-through target page configured with University Name filter parameter.	Right-clicking any institution in Top 10 or Scatter plots opens deep individual institutional profile.
Theme Customization	Custom JSON Theme (`EduVision_Theme.json`) loaded into Power BI Desktop.	Applies universal dark canvas `#0B0E1B`, card containers `#1B1F38`, Segoe UI font, and high-contrast color palette.
Data Model	Star Schema: `University_Performance` (Fact Table) joined to `Regional_Benchmarks` and `Country_Benchmarks`.	High query efficiency with single-direction 1:* relationships and dedicated `DAX_Measures` table.